

non-Hausdorffness

This is a quick blog to explain and construct a non-Hausdorff spaces to someone with no university background. There shall be some small apocrypha's for the sake of not overloading on technicalities, and these shall be sorted in an extra technicalities section.

1 Build up

The set of all numbers that are not complex are called the *real number line* and is denoted $\mathbb{R}$. It is often drawn as:

$$\mathbb{R} \quad \cdots \quad \begin{array}{c} | \quad | \quad | \quad | \quad | \quad | \\ -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3 \end{array} \quad \cdots$$

An element of $\mathbb{R}$ is called a *real number* (to distinguish from complex numbers). Let us choose any two real numbers, say 3 and 4. Then they are a distance 1 apart from each other. Any two *distinct* real numbers x, y have some distance, namely $x - y$ or $y - x$ (depending on which number is bigger). If the distance between two real numbers is 0, then they are the same number, since:

$$x - y = 0 \quad \text{then} \quad x = y$$

But there are spaces in which you can have two points that have *zero distance* between them and yet are *distinct*! It was a mathematician named *Hausdorff* that first stumbled upon such a weird space, and thus in his honour a space in which there can exist distinct points that have zero distance are called *non-Hausdorff spaces* and a space is called *Hausdorff* otherwise. For example the real numbers $\mathbb{R}$ is Hausdorff.

2 What is a Space?

What is meant here by “space”? The first encounter we all have with the notion of “space” is within the world we live in: we consider the “space” of a room, or the routes of a hiking trail the “space” on which we can walk. These two examples of spaces are very different from each other (in a room, we have 3 directions, while for hiking trails we usually are limited to lines to follow), and there are many more such examples that come up in the sciences (for example, we consider all possible outputs of an experiment the “total space” of outcomes). This generality of the notion of space has lead mathematics to define the notion of *topology* which is a vast generalization of the Cartesian plane. Going through the technicalities of defining a topology would distract from the specific goal of constructing a non-Hausdorff space, so we shall define space as:

A set containing numbers and sets of numbers

This is very abstract, so let us immediately justify this definition. Let us start by being precise what we mean by *set*. A set is simply the way mathematicians group things, for example here are two different sets, one containing the numbers 1, 2, 3 and the other the numbers 3, 4, 5:

$$\{1, 2, 3\} \quad \text{and} \quad \{3, 4, 5\}$$

where we used the $\{, \}$ notation to enclose the elements we are considering. Sets can have more than numbers, for an arbitrary example a set can contain all the words in the English dictionary:

$$S = \{\dots, \text{Hamiltonian}, \text{characteristic}, \text{Hypothesis}, \dots\}$$

A set can also contain other sets (think by analogy a warehouse that has boxes containing stuff):

$$W = \{\{\text{Alice in Wonderland}, \text{Introduction to Analysis}\}, \{\text{spoon}, \text{fork}, \text{knife}\}\}$$

We can also have sets containing numbers and sets of numbers, for example:

$$R = \{1, 2, \{3, 4\}\}$$

This is exactly what we are calling a space. Next it must be justified why numbers and sets of numbers are being mixed. The idea is that the numbers should be considered as points, and the sets of numbers should be thought of as being points that are *glued together*.

Let us demonstrate this with an example: the following set splits the even and the odd numbers into two sets:

$$P = \{\{0, 2, 4, 6, \dots\}, \{1, 3, 5, 7, \dots\}\} = \{E, O\}$$

where we labeled the sets within P as E for even numbers and O for odd numbers. Notice that P only has two sets (naturally each set contains infinitely elements). What we shall do now is show how we can interpret the fact that P has 2 sets as all the elements within E and O respectively “equal” to each other, that is:

$$0 = 2 = 4 = \dots \quad \text{and} \quad 1 = 3 = 5 = \dots$$

Here is how to think of this equality: the set P *only* remembers the *parity* of each number. It does not care if a number is bigger or smaller or divisible by 5, it only cares if it is even or odd. The equality in the above equation is an *equality of parity*, meaning two numbers are the same if they have the same parity. Using the equal sign “=” is confusing as we already use it to mean to numbers are “exactly equal” not just “equal in parity” (or as a mathematician would say, equal *up to parity*). Let us thus create a new symbol $\sim_P$ and write:

$$0 \sim_P 2 \sim_P 4 \sim_P \dots \quad \text{and} \quad 1 \sim_P 3 \sim_P 5 \sim_P \dots$$

So $a \sim_P b$ is both a and b are even, or both a and b are odd, and otherwise we write $a \not\sim_P b$. We can go one step further and say that we “glued” these numbers together. If we represent the numbers $\{0, 1, 2, \dots\}$ as points we get:

$$\begin{array}{cccccccccc} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \dots \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \end{array}$$

and the set P can be represented as:

$$\begin{array}{cc} \bullet & \bullet \\ E & O \end{array}$$

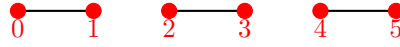
Then the set P glued all the even points together and all the odd points together.

Thinking of the above process as gluing is actually how mathematicians construct many objects. To show this is not just a trick, let us show how we can create a triangle using this method. Say we had three straight lines L_1, L_2, L_3 each of length 1. Let us represent each as intervals:

$$\begin{aligned} L_1 &= [0, 1] = \{x \text{ in } \mathbb{R} \text{ such that } 0 \leq x \leq 1\} \\ L_2 &= [2, 3] = \{x \text{ in } \mathbb{R} \text{ such that } 2 \leq x \leq 3\} \\ L_3 &= [4, 5] = \{x \text{ in } \mathbb{R} \text{ such that } 4 \leq x \leq 5\} \end{aligned}$$

2. What is a Space?

Notice that we are taking x in the real numbers; for example in L_1 all numbers between 0 and 1 (including 0 and 1) are in L_1 (ex. 0.5 and $\pi - 3$ are in L_1). Visually, these three sets represent lines of length 1:



Let us now glue these lines to form a triangle, we want to:

1. glue 1 in L_1 to 2 in L_2
2. glue 3 in L_2 to 4 in L_3
3. glue 5 in L_3 to 0 in L_1

To do this, we shall introduce a bit of notation. Let us separate the end-points of the lines so that we may glue them. Let:

$$(0, 1) = \{x \text{ in } \mathbb{R} \text{ such that } 0 < x < 1\}$$

$$(2, 3) = \{x \text{ in } \mathbb{R} \text{ such that } 2 < x < 3\}$$

$$(4, 5) = \{x \text{ in } \mathbb{R} \text{ such that } 4 < x < 5\}$$

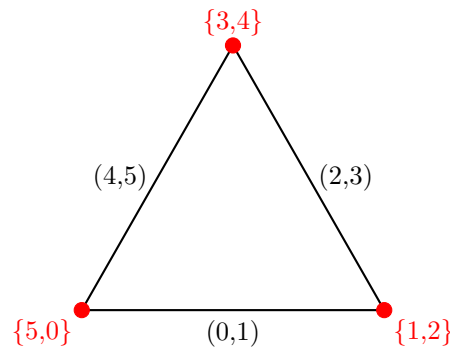
Notice the strict inequalities now! We now want to combine the points representing the straight-edges of a triangle with the sets representing the vertices of a triangle. To do this, we take the *union* of sets. The union of sets is represented by $\cup$ and is written as:

$$A \cup B = \{\text{take all } a \text{ in } A \text{ and all } b \text{ in } B\}$$

In other words, $A \cup B$ is a new set that containing all the elements of A and B . Define our triangle to be:

$$T = (0, 1) \cup \{\{1, 2\}\} \cup (2, 3) \cup \{\{3, 4\}\} \cup (4, 5) \cup \{\{5, 0\}\}$$

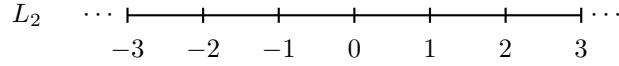
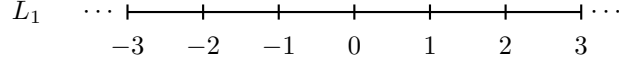
Let us break this down. T contains each element inside $(0, 1)$, $(2, 3)$ and $(4, 5)$. It then also contains each element inside the three sets $\{\{1, 2\}\}$, $\{\{3, 4\}\}$, $\{\{5, 0\}\}$. Each of these sets contains only a single element: they contain $\{1, 2\}$, $\{3, 4\}$, $\{5, 0\}$ respectively that represents gluing the vertices of these lines together. We can visualize T as:



Giving us our Triangle!

Non-Hausdorff Space

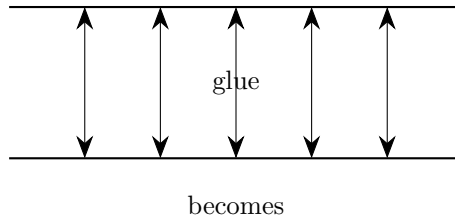
With this, we can finally construct our non-Hausdorff space. The construction is in fact rather simple. Let L_1 and L_2 be two copies of the real number line:



To not get confused with which numbers are we working with, let us label all numbers x in L_1 as x_1 and all numbers x in L_2 as x_2 , for example 3_1 is in L_1 and 3_2 is in L_2 . Let us now glue *all* the numbers x_1 in L_1 to the number x_2 in L_2 that has the same value, *except* for 0_1 and 0_2 . So take all sets of numbers:

$$\{x_1, x_2\} \text{ (ex. } \{1_1, 1_2\}, \{32_1, 32_2\}, \dots) \text{ but } 0_1, 0_2 \text{ are not in a set}$$

Taking the union of all these sets, we get a space N (N for “non-Hausdorff”). Let us think about this set for a moment. for any non-zero number, we glued them together:



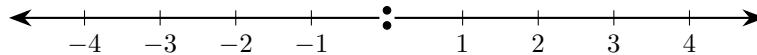
Thus, we don't need to distinguish between the nonzero points in N : we can just say 3 is in N as we glued $\{3_1, 3_2\}$. To be precise, we can label the set $\{3_1, 3_2\}$ as 3, and more generally if $x \neq 0$ we can say x is in N instead of $\{x_1, x_2\}$ is in N . As we have not glued the points 0_1 and 0_2 , they remain distinct, so 0_1 and 0_2 are in N .

Now, we can finally ask:

What is the distance between 0_1 and 0_2 ?

We shall show they have 0 distance between each other. Let's say that there *was* some distance d between these points. Then we can find a nonzero number t that is between 0_1 and 0_2 and distance $d/2$ from either 0_1 or 0_2 . But that is absurd; there is *no* number between the zero's. Namely, the moment we try to move left from 0_1 or 0_2 , we are encountering a negative number that is left from both numbers, and if we move right from 0_1 or 0_2 , we are encountering a positive number that is right from both numbers, so there can't be a number between 0_1 or 0_2 . But then they have no distance separating them!

The space N is often attempted to be visualized as:



However this is misleading, as it looks like there is a gap between 0_1 and 0_2 , but there is not. It is impossible to visualize this space as we live in a Hausdorff world¹. Nevertheless, this image can be

¹At least on the macro scale we live in

used to ground the intuition that there is no gap between 0_1 and 0_2 ; two points in the middle can be thought of as existing “one atop of another” in the same location.

Though this might seem like an exercise in arm-chair mathematising, these spaces do show up naturally in mathematics, especially within number theory and algebraic geometry. Unfortunately, to see why they show up *naturally* takes a lot of build up, and presenting them without this build-up will look a lot like random ideas stuck together. But this is also not too surprising as it took thousands of years before such spaces were found/invented, and then to realise that such spaces have immense practical properties in the 1950s (very recent mathematics!).

3 Technicalities

Defining a space as the set of numbers and subset of numbers is too vague; It doesn't define any notion of “locality” that is the key notion in topology, namely we have no notion of open or closed sets, and there are no restrictions on which sets can be part of a space (ex. we don't want the countable intersection of open sets in our collection) Even worse, as every tuple is technically a set, the definition of space presented above includes $\mathbb{R}^2, \mathbb{R}^3, \dots$, and in some model's of math can include any sets as all these space are ultimately sets of numbers, but it is “evident” that the definition is really only considering one-dimensional spaces. The first step to fixing this is by defining a topological space as a set of sets containing the empty set and itself, is closed under finite intersection, and arbitrary unions. We now have a notion of locality, and can start adding further constrictions such as being metrizable to actually have distance be well-defined. All such details can be found in [2].

The Hausdorff condition is formally defined as:

Given two points $x, y \in X$ in a topological space X , there exists open sets U_x, U_y containing x, y respectively such that $U_x \cap U_y = \emptyset$

This shows that points can be *separated*. The approach taken here with the idea of distance is to avoid introducing the more abstract notion of “separation” and stick to the intuitive notion of distance.

The examples of glued sets almost respect the rules of an equivalence relationship; we technically should separate each number into a singleton set so that our space is a set of equivalence classes. The details can be found in [1].

References

- [1] Nathanael Chwojko-Srawley. *Preliminary Chapter*. 2023. URL: /Users/nathanaelchwojko-srawkey/website-nate/nate-website/src/assets/latex/notes/preliminary_chapter.
- [2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Topology*. 1st ed. 2022. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_topology.pdf.